Syntax:

datatype * variable; /* **ptr declaration** */

Here * indicates it is a pointer type variable.

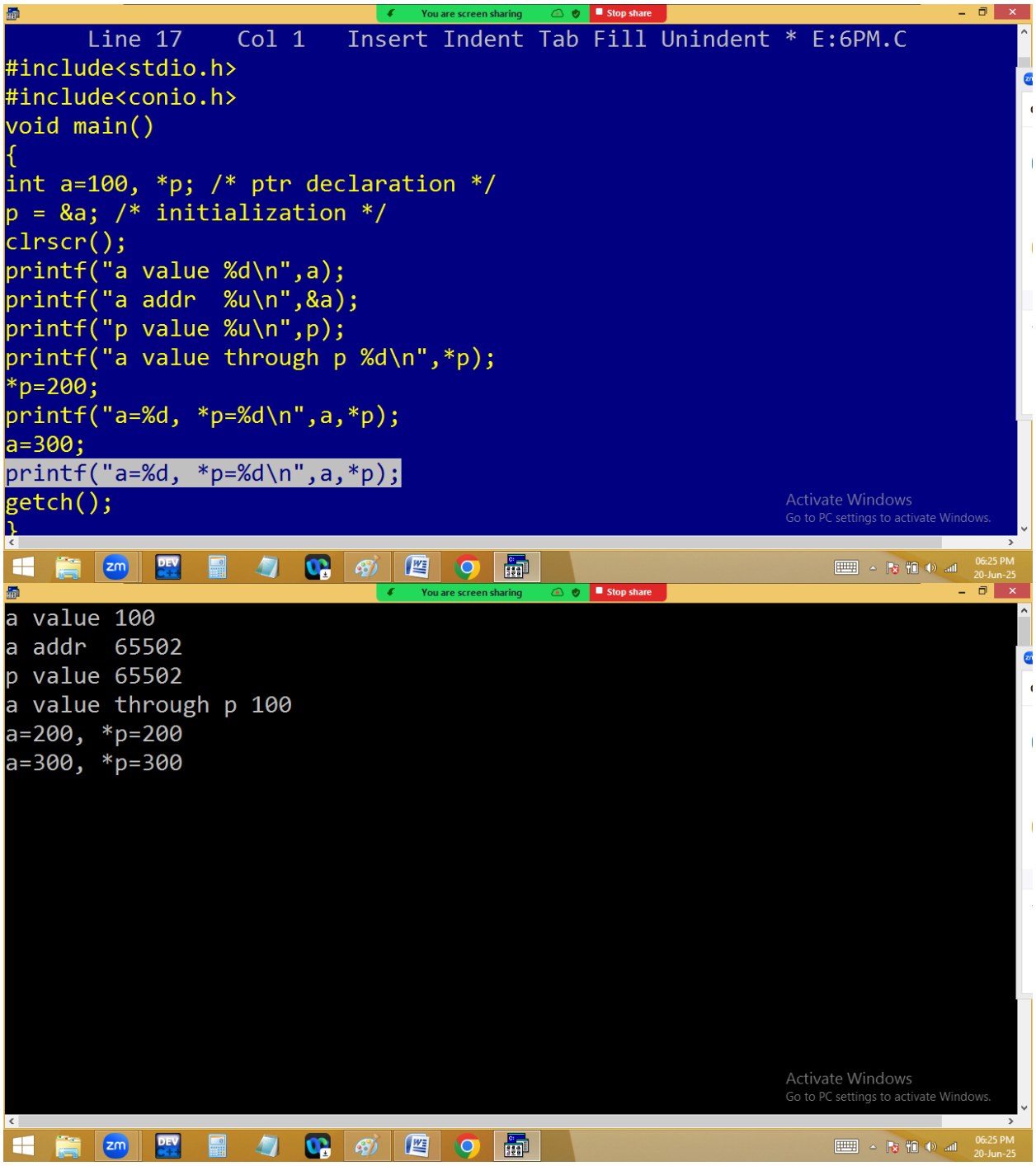
Eg:

```
int a=100; /* local var declaration & initialization */

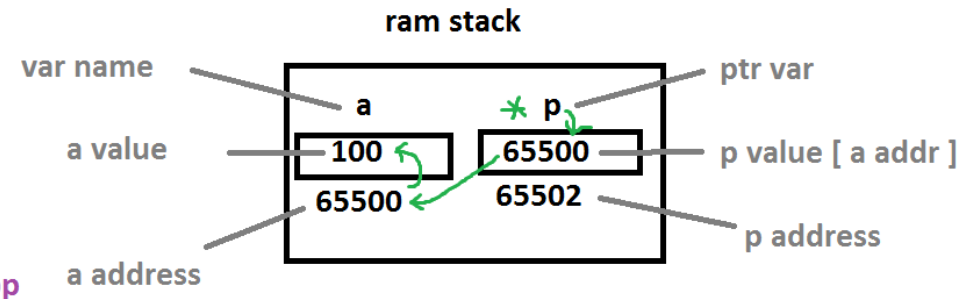
int *p;  /* ptr var declaration */

p = &a;  /* ptr var initialization */
```

Finding a variable value and address using pointer:


You are screen sharing
Stop share
Line 17 Col 1 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=100, *p; /* ptr declaration */
p = &a; /* initialization */
clrscr();
printf("a value %d\n",a);
printf("a addr %u\n",&a);
printf("p value %u\n",p);
printf("a value through p %d\n",*p);
*p=200;
printf("a=%d, *p=%d\n",a,*p);
a=300;
printf("a=%d, *p=%d\n",a,*p);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:25 PM
20-Jun-25
You are screen sharing
Stop share
a value 100
a addr 65502
p value 65502
a value through p 100
a=200, *p=200
a=300, *p=300
Activate Windows
Go to PC settings to activate Windows.
06:25 PM
20-Jun-25

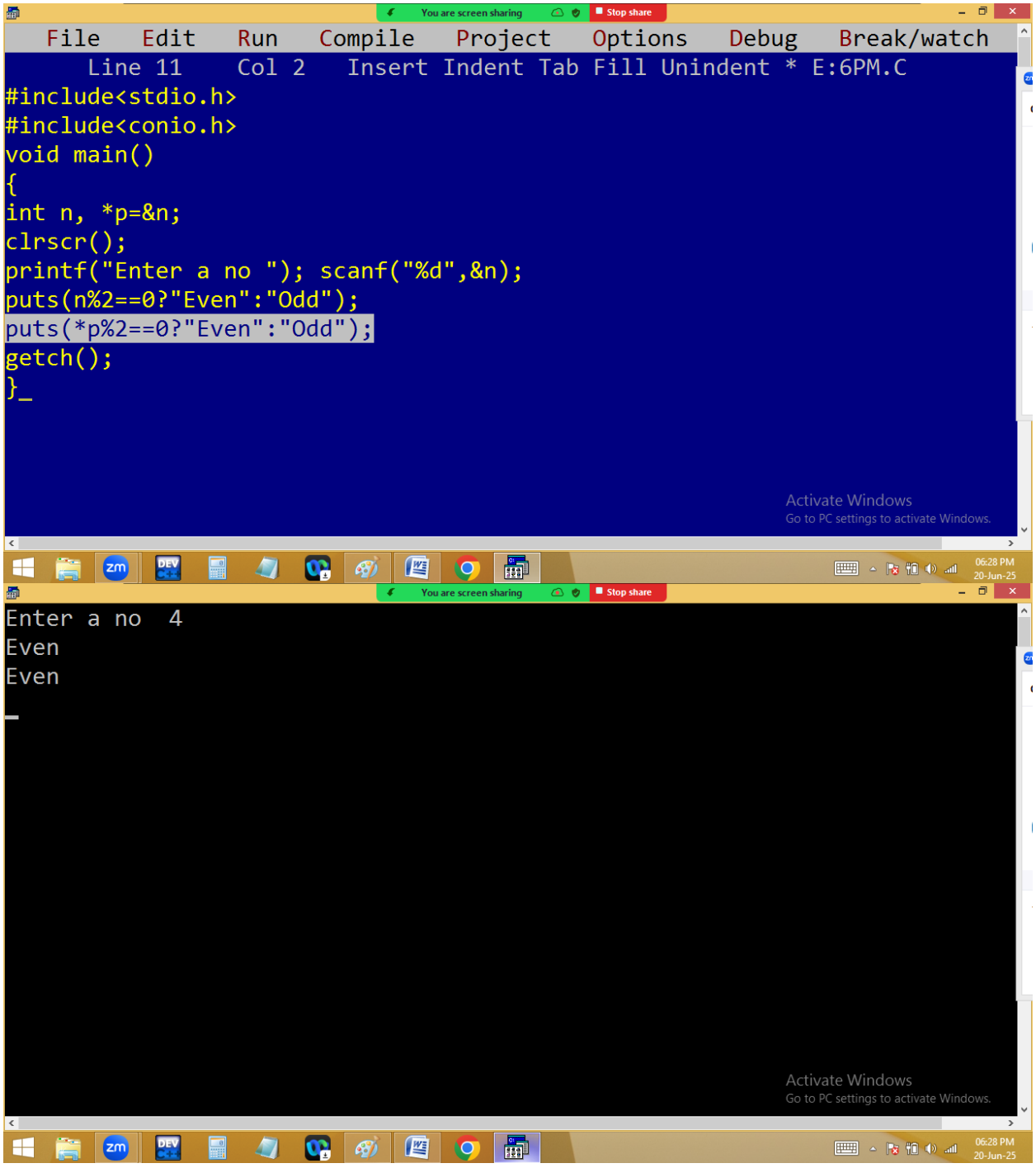
```
int a=100;
int *p = &a;
```

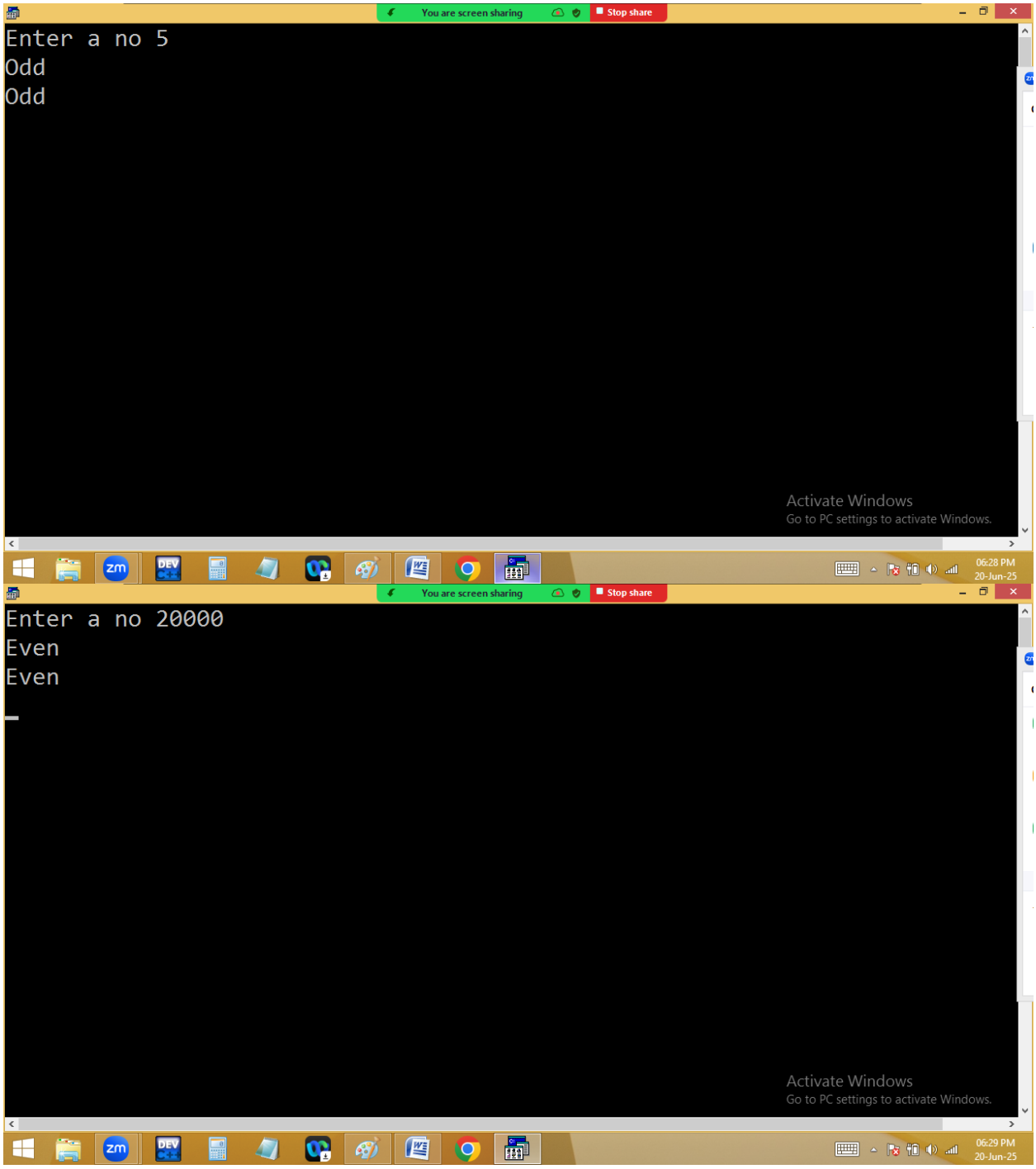


* is called:
indirection op
redirection op
dereferencing op
value in that addr
value of ptr

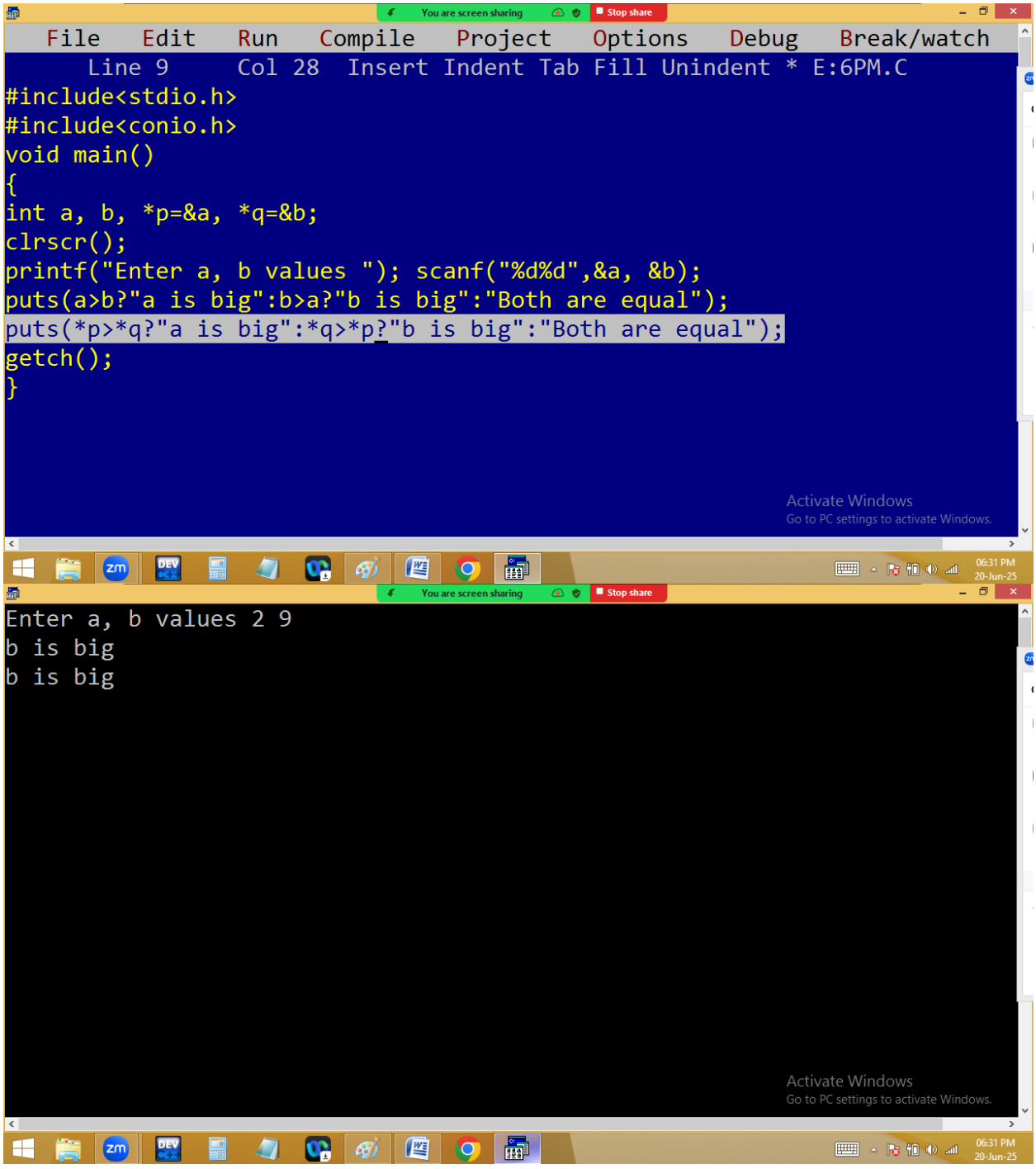
printf(" %d", * p); ==> 100
↳ 65500
↳ *65500 ==> value at 65500 ==> 100

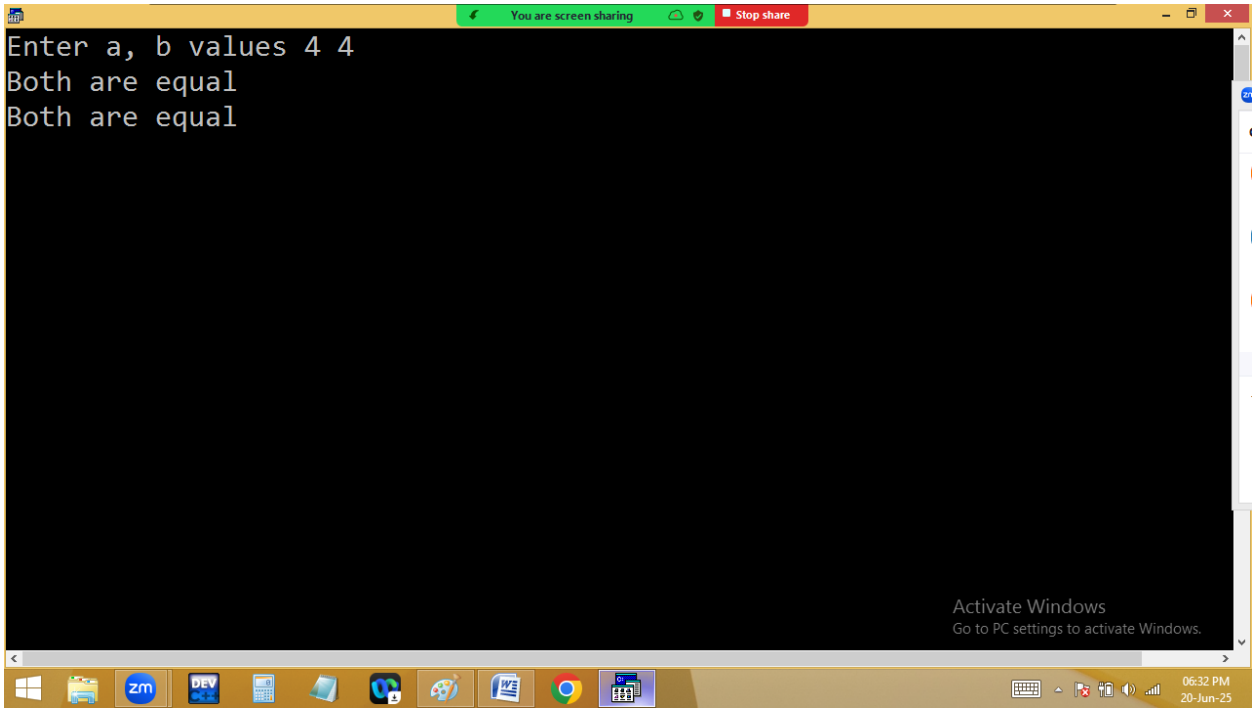
Finding even/odd using a pointer:


You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 2 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n, *p=&n;
clrscr();
printf("Enter a no "); scanf("%d",&n);
puts(n%2==0?"Even":"Odd");
puts(*p%2==0?"Even":"Odd");
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:28 PM
20-Jun-25
You are screen sharing
Stop share
Enter a no 4
Even
Even
Activate Windows
Go to PC settings to activate Windows.
06:28 PM
20-Jun-25

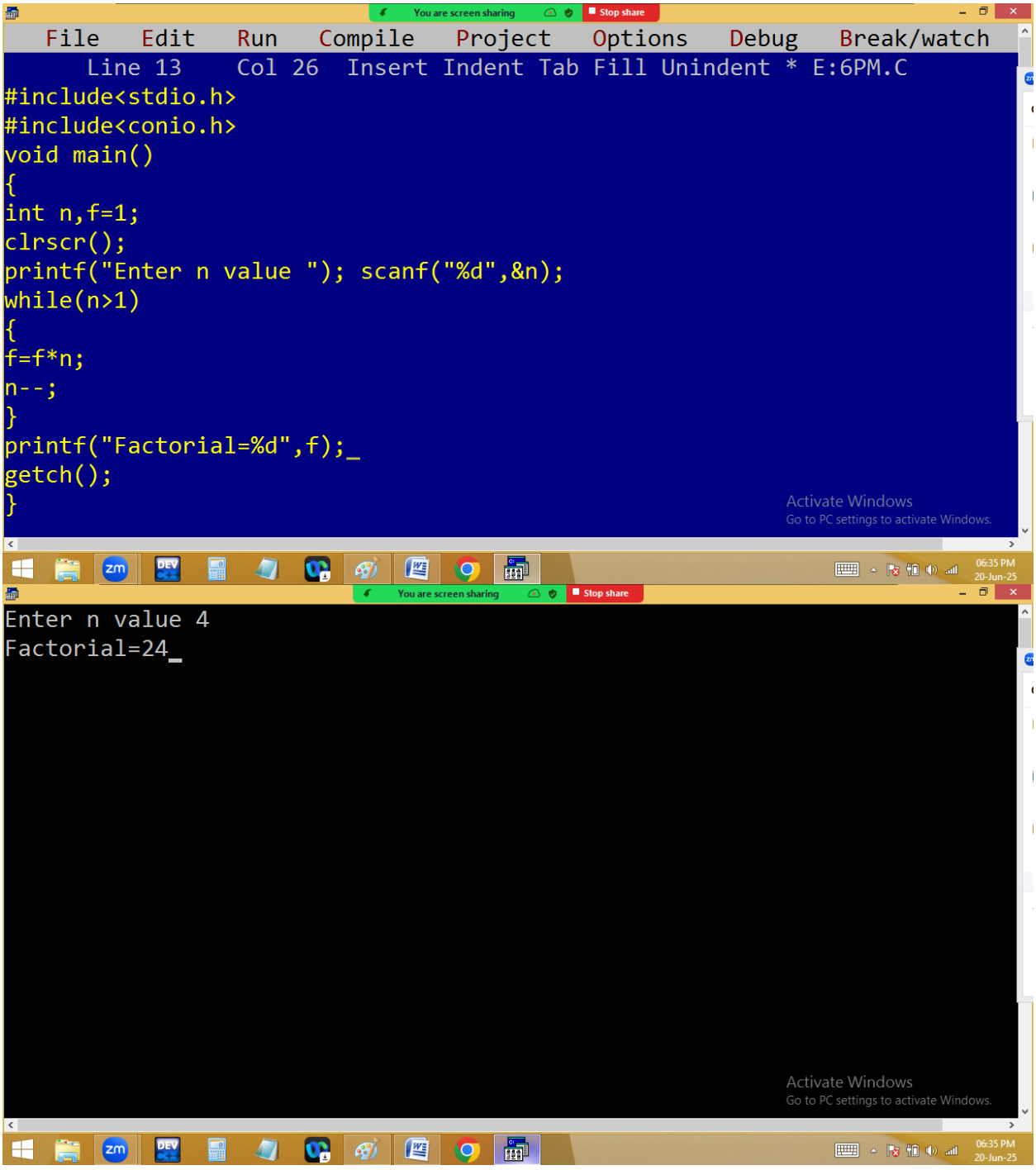


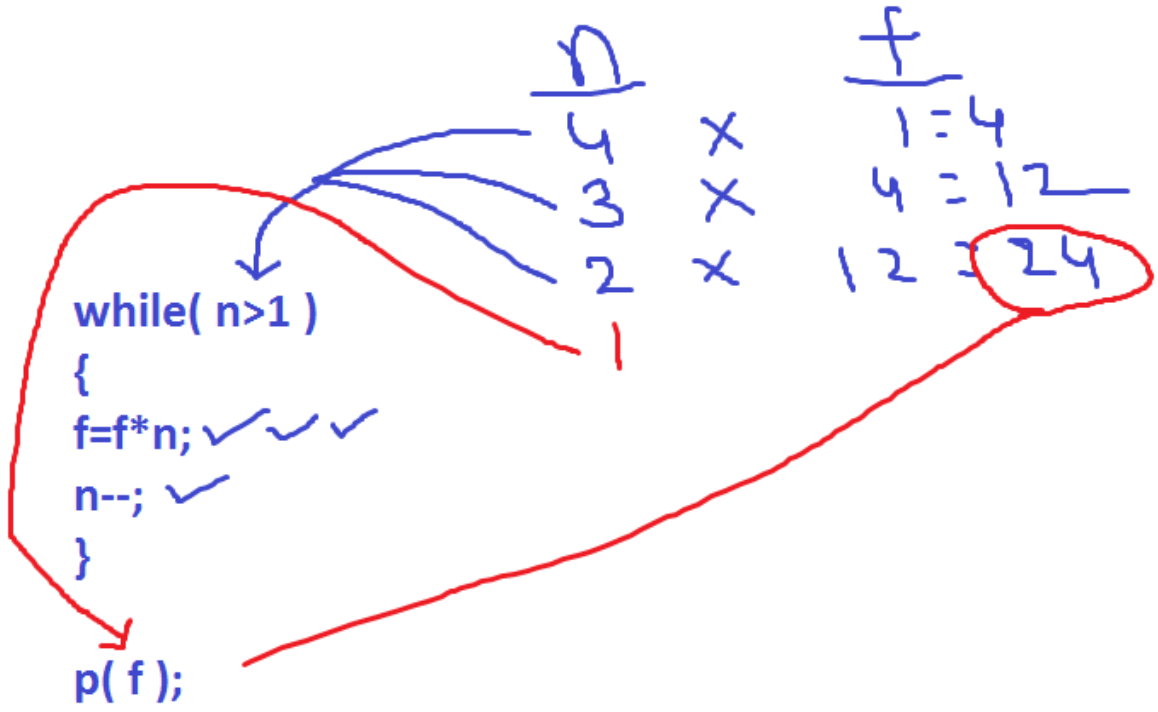
Finding max in 2 no's using of pointer?


You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 28 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a, b, *p=&a, *q=&b;
clrscr();
printf("Enter a, b values "); scanf("%d%d",&a, &b);
puts(a>b?"a is big":b>a?"b is big":"Both are equal");
puts(*p>*q?"a is big":*q>*p?"b is big":"Both are equal");
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:31 PM
20-Jun-25
You are screen sharing
Stop share
Enter a, b values 2 9
b is big
b is big
Activate Windows
Go to PC settings to activate Windows.
06:31 PM
20-Jun-25

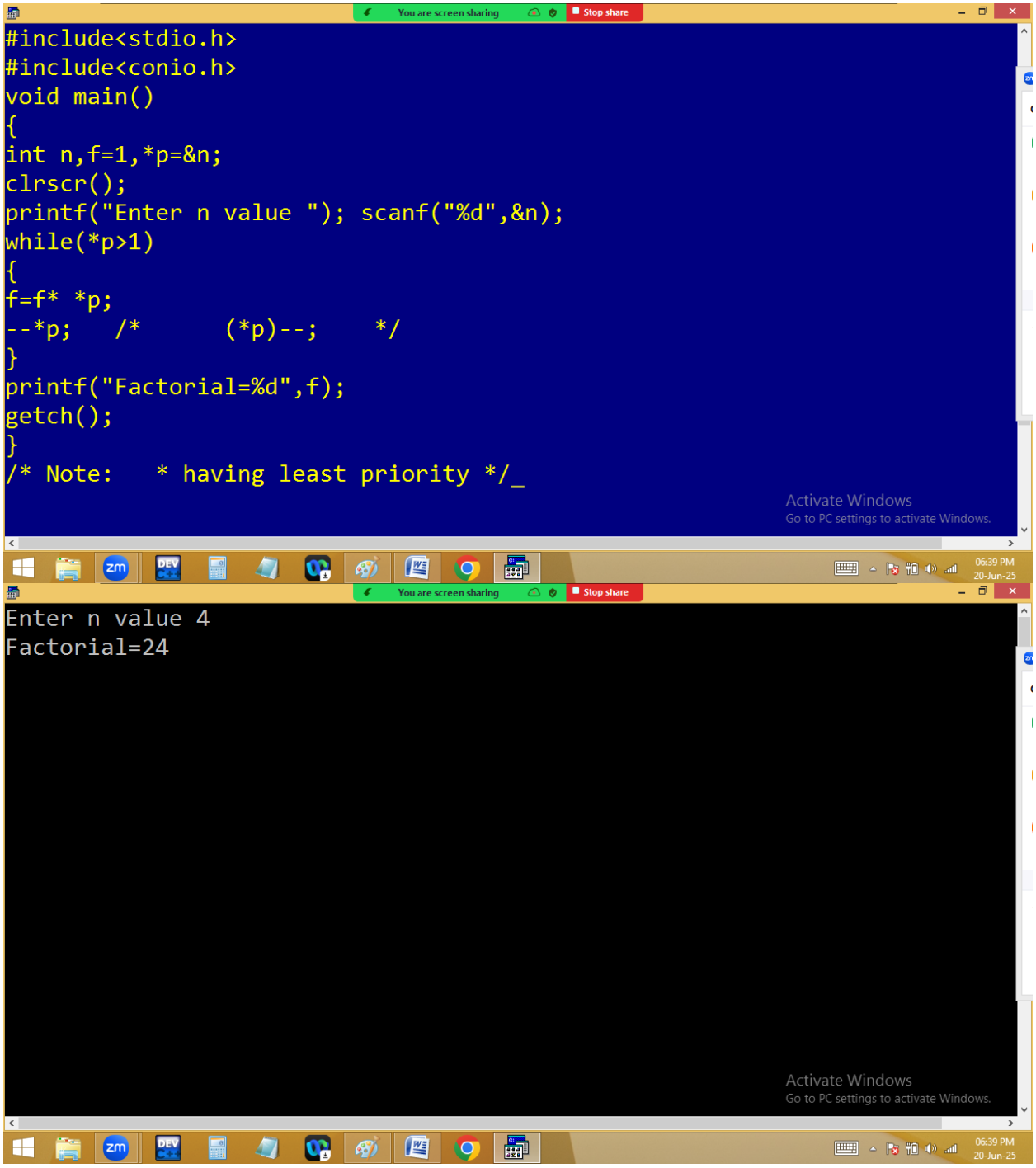


Finding factorial of given no using pointer?

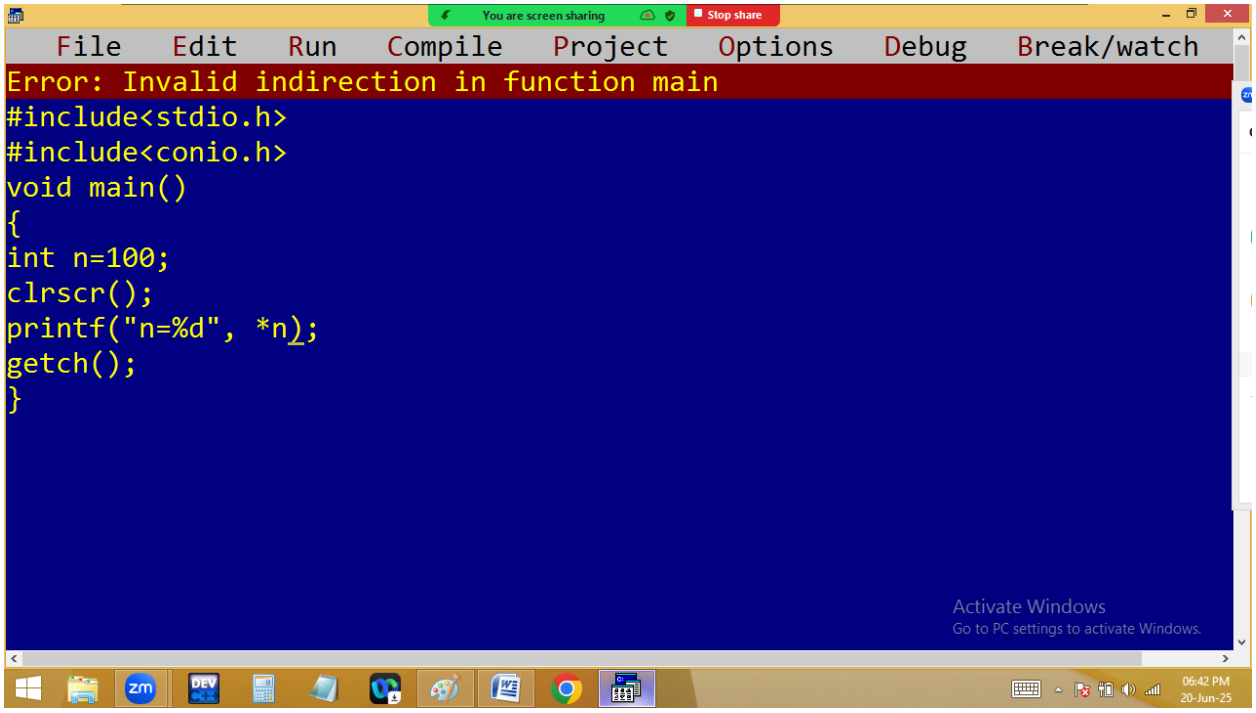

You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 26 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n,f=1;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(n>1)
{
f=f*n;
n--;
}
printf("Factorial=%d",f);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:35 PM
20-Jun-25
You are screen sharing
Stop share
Enter n value 4
Factorial=24
Activate Windows
Go to PC settings to activate Windows.
06:35 PM
20-Jun-25

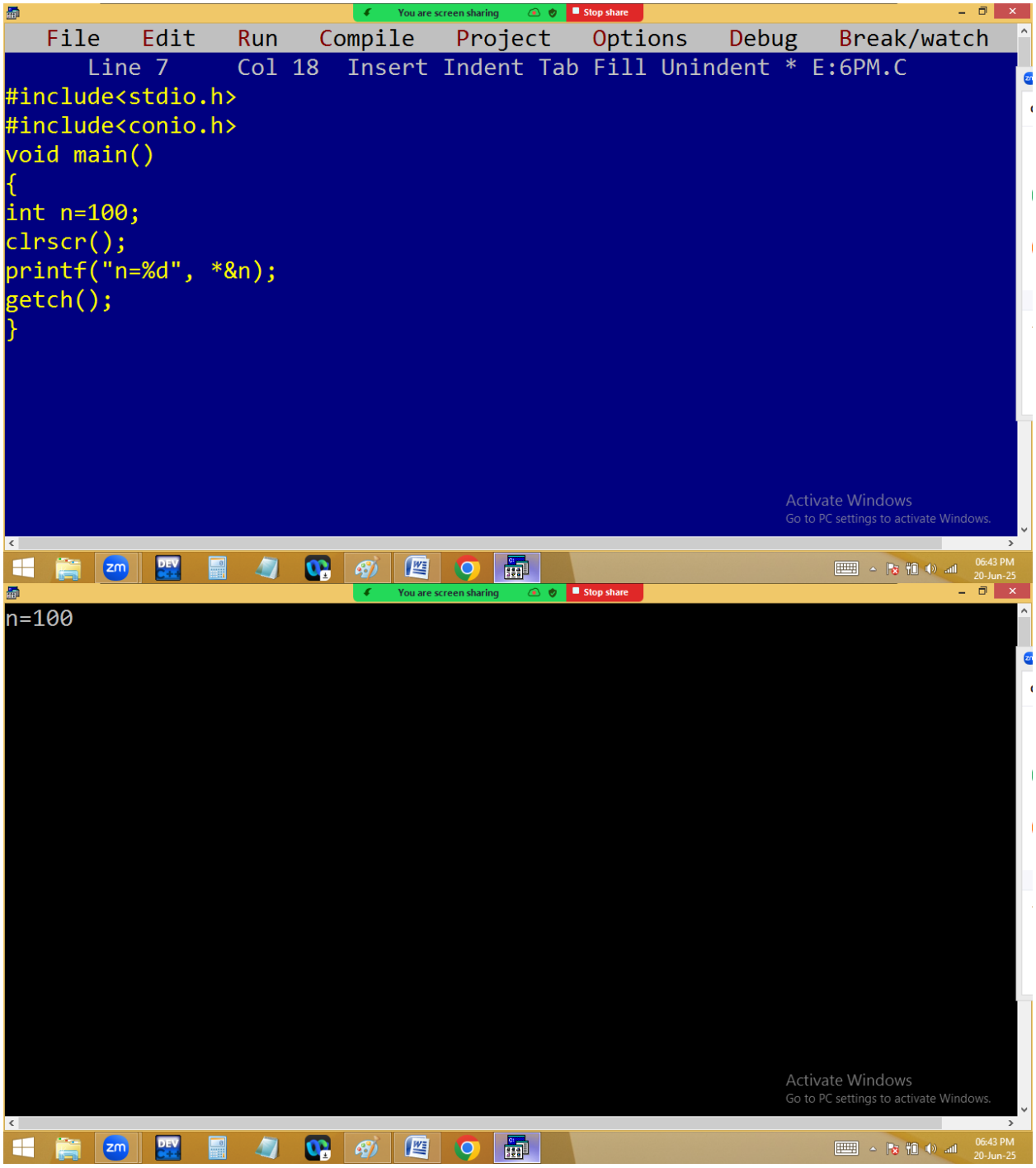


Using pointer:



Write a c program to print a normal variable value using pointer technique?

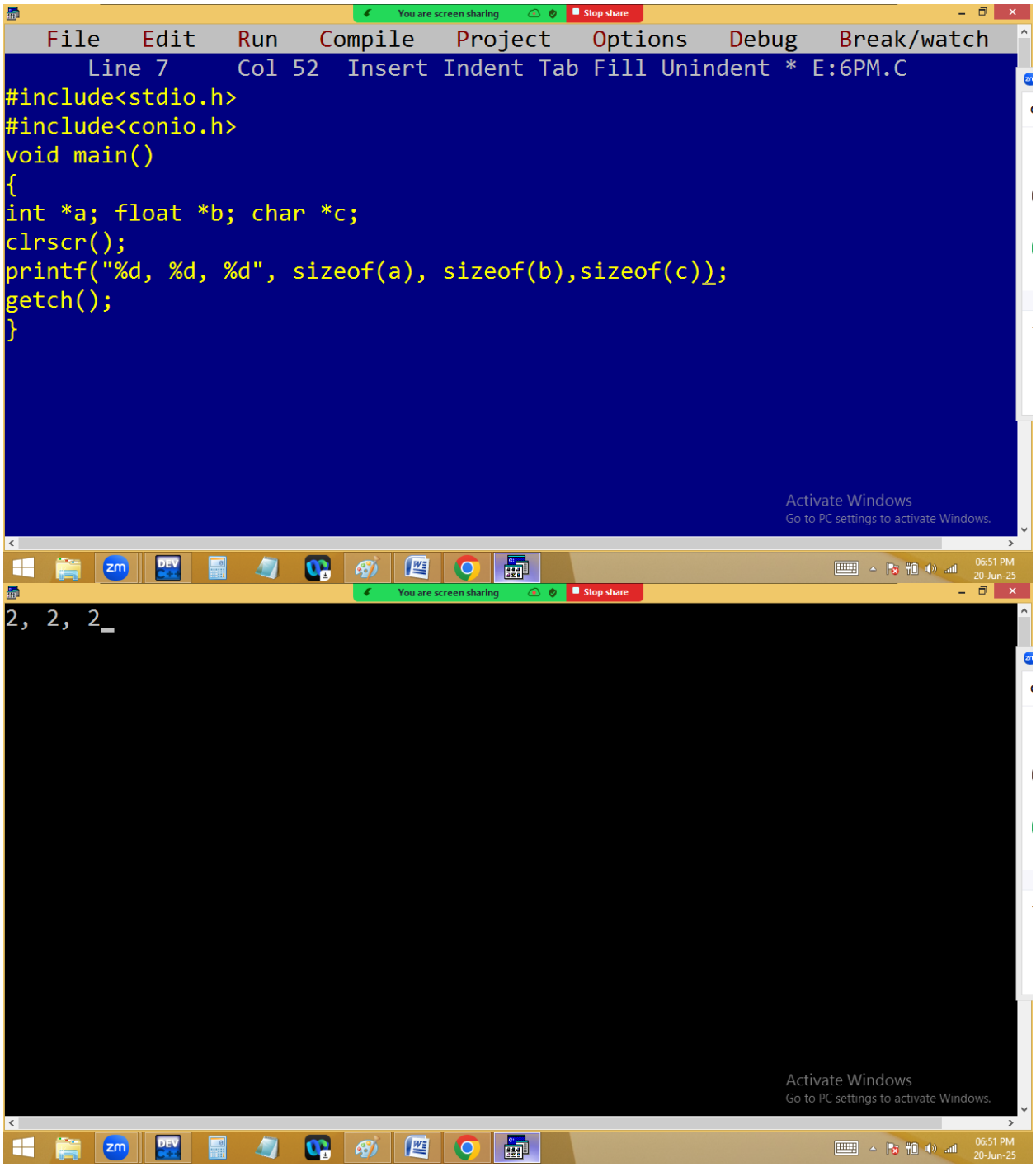

You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Error: Invalid indirection in function main
#include<stdio.h>
#include<conio.h>
void main()
{
int n=100;
clrscr();
printf("n=%d", *n);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:42 PM
20-Jun-25

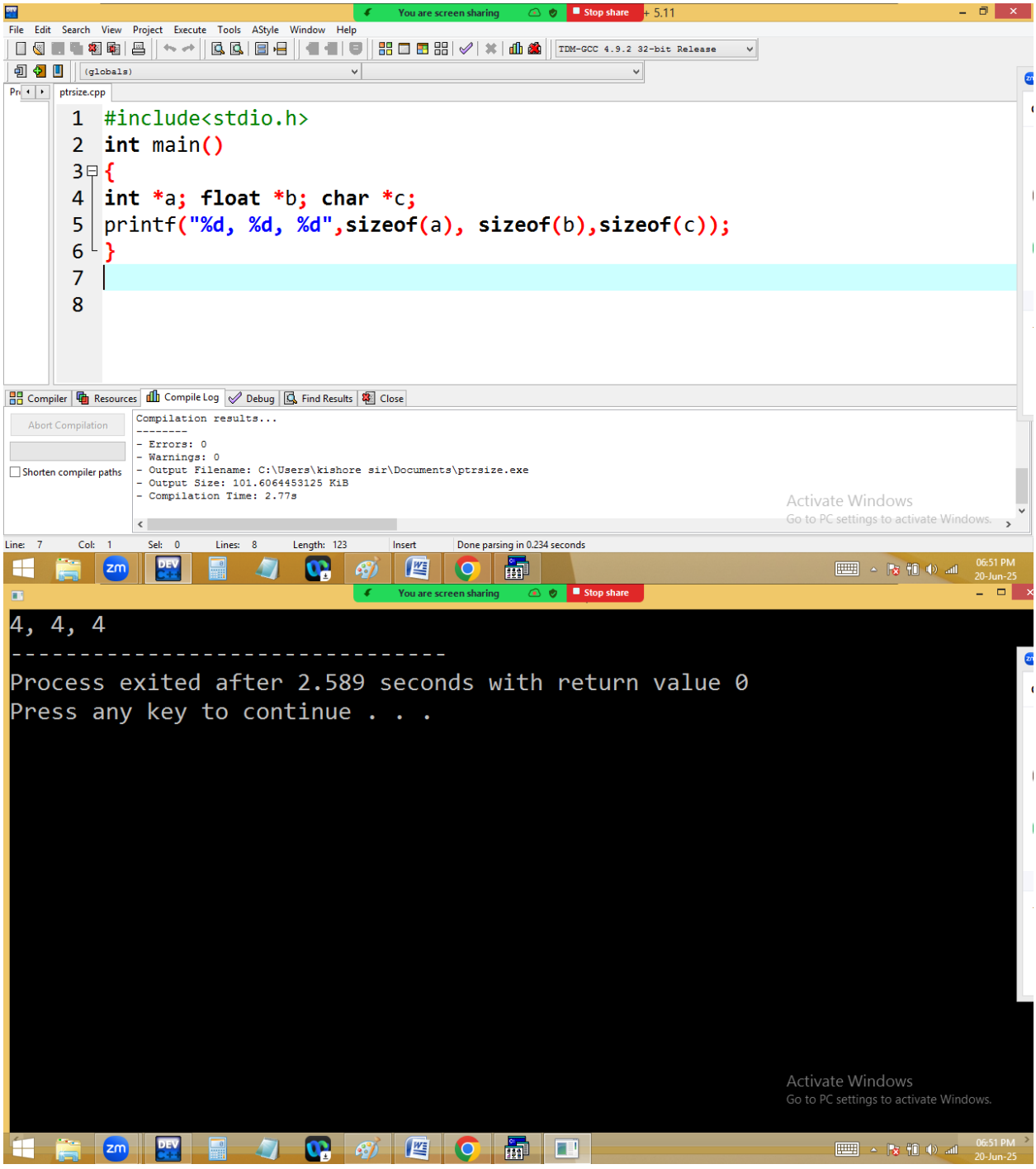


Finding pointer size:

Any type of pointer it stores only the address and address is always unsigned int. due to this the

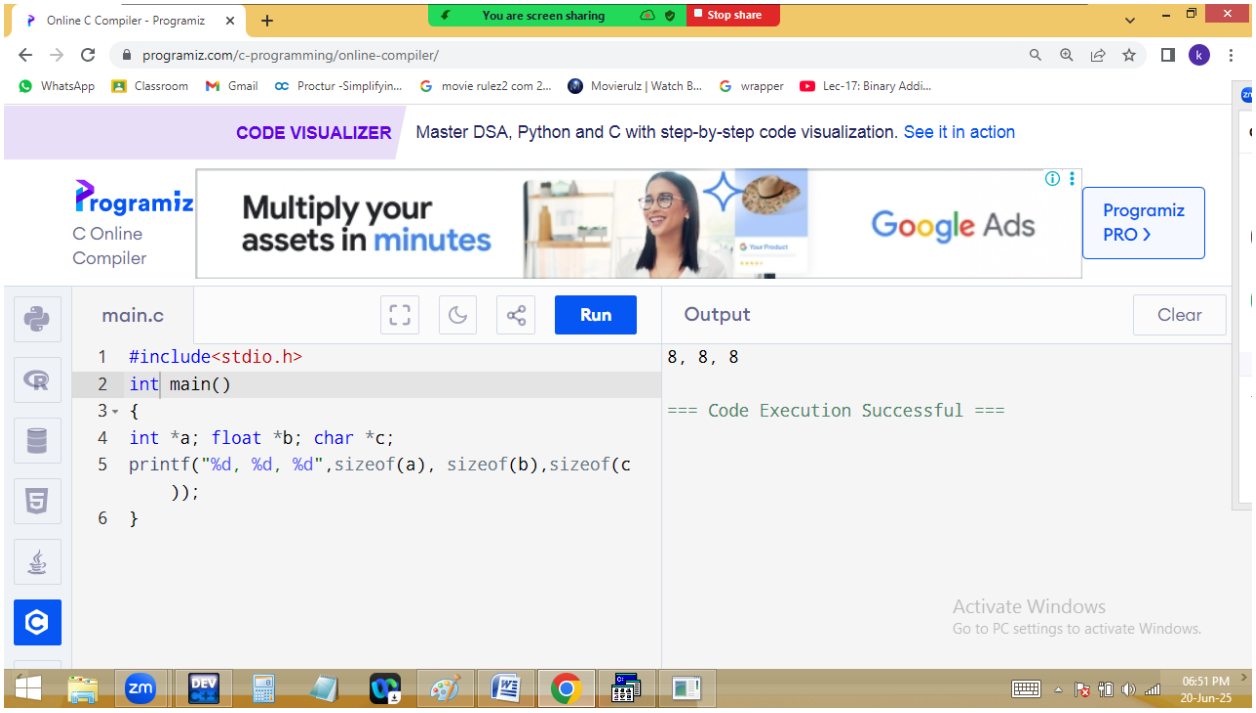
pointer takes 2 / 4 / 8 bytes in 16 / 32 / 64 bit compilers.



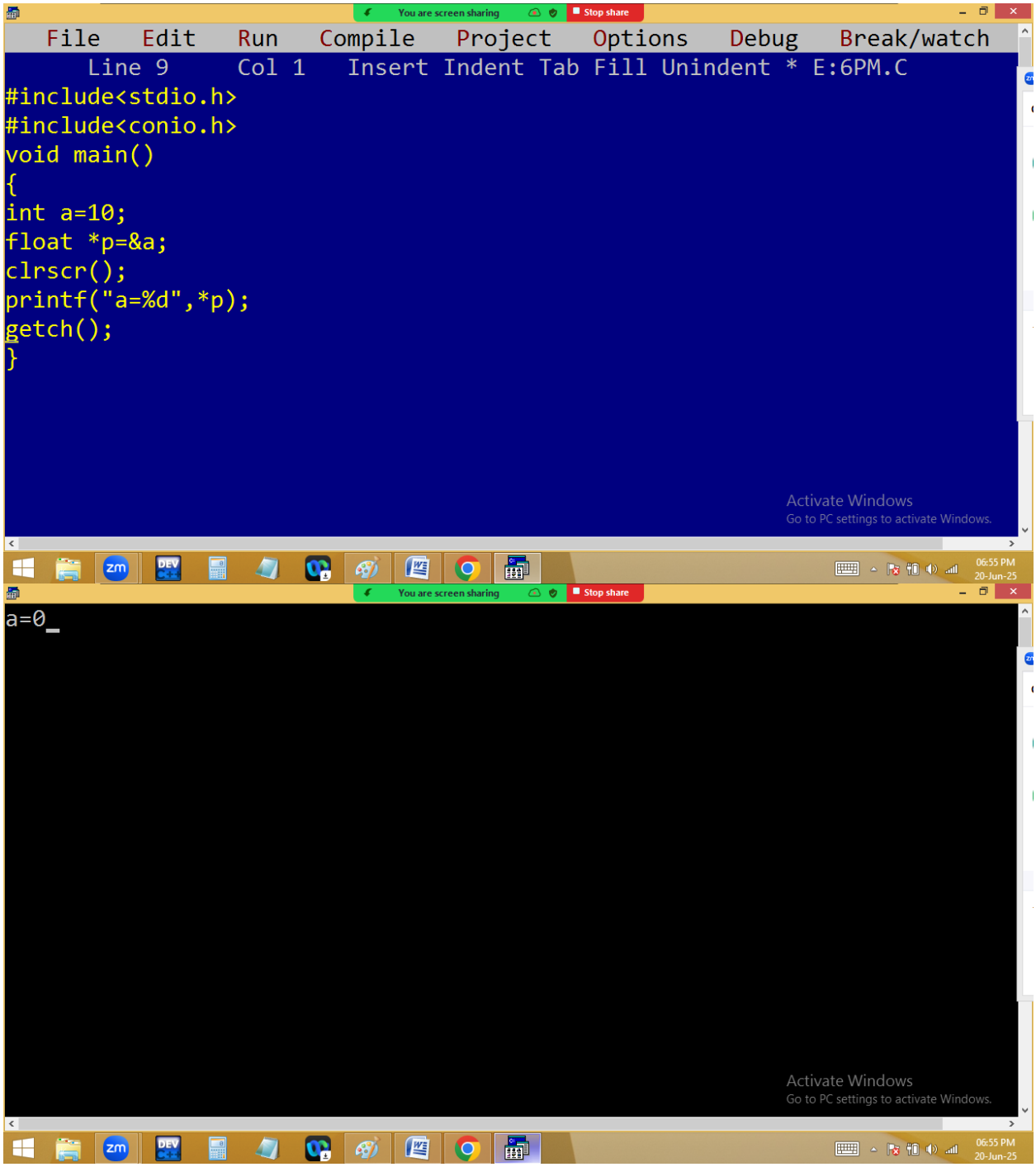

You are screen sharing
Stop share
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 32-bit Release
(globals)
ptrsize.cpp
#include<stdio.h>
int main()
{
int *a; float *b; char *c;
printf("%d, %d, %d",sizeof(a), sizeof(b),sizeof(c));
}
Compiler Resources Compile Log Debug Find Results Close
Abort Compilation
Shorten compiler paths
Compilation results...

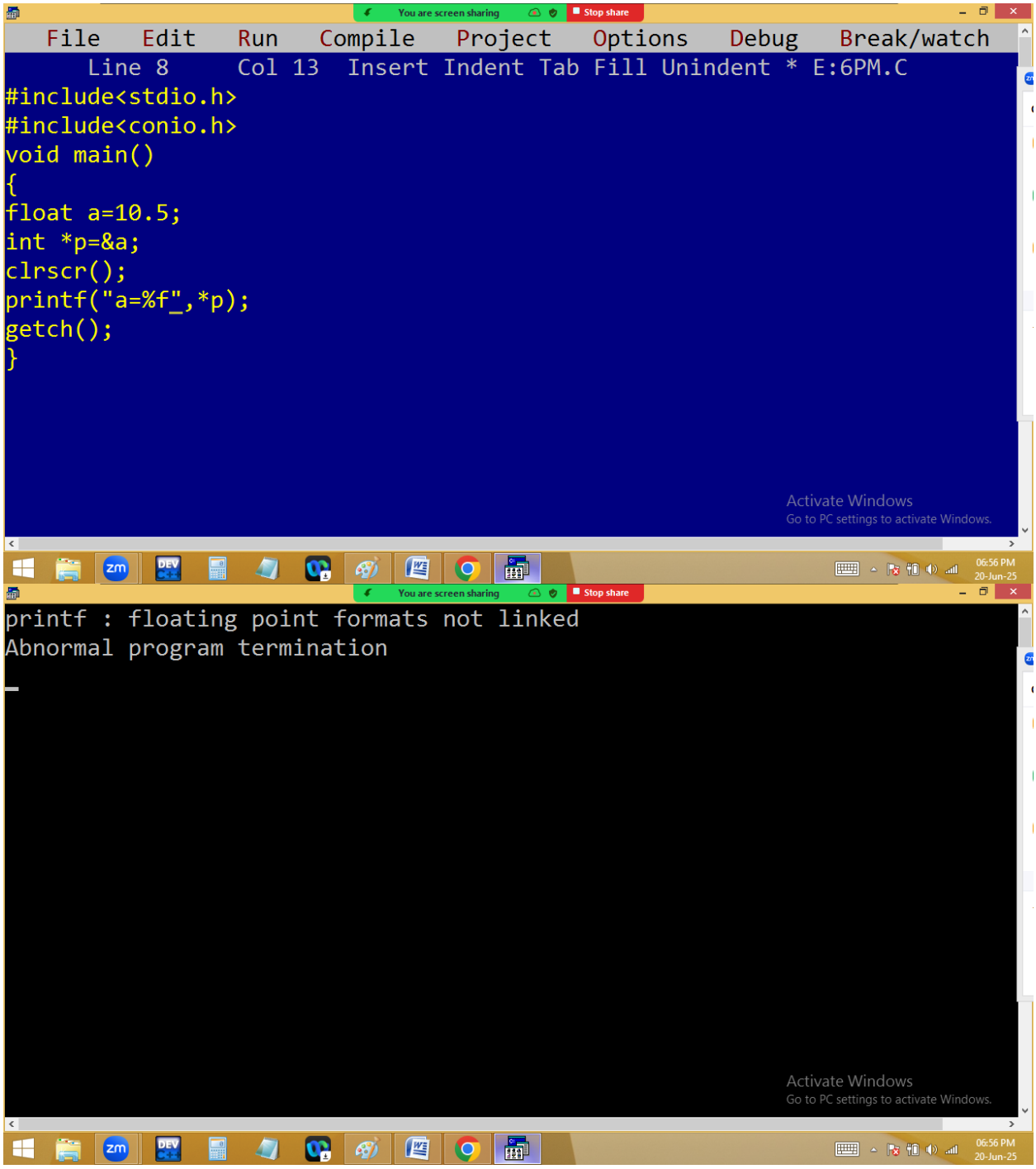
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\kishore sir\Documents\ptrsize.exe
- Output Size: 101.6064453125 KiB
- Compilation Time: 2.77s
Activate Windows
Go to PC settings to activate Windows.
Line: 7 Col: 1 Sel: 0 Lines: 8 Length: 123 Insert Done parsing in 0.234 seconds
06:51 PM
20-Jun-25
You are screen sharing
Stop share
4, 4, 4

Process exited after 2.589 seconds with return value 0
Press any key to continue . . .
Activate Windows
Go to PC settings to activate Windows.
06:51 PM
20-Jun-25



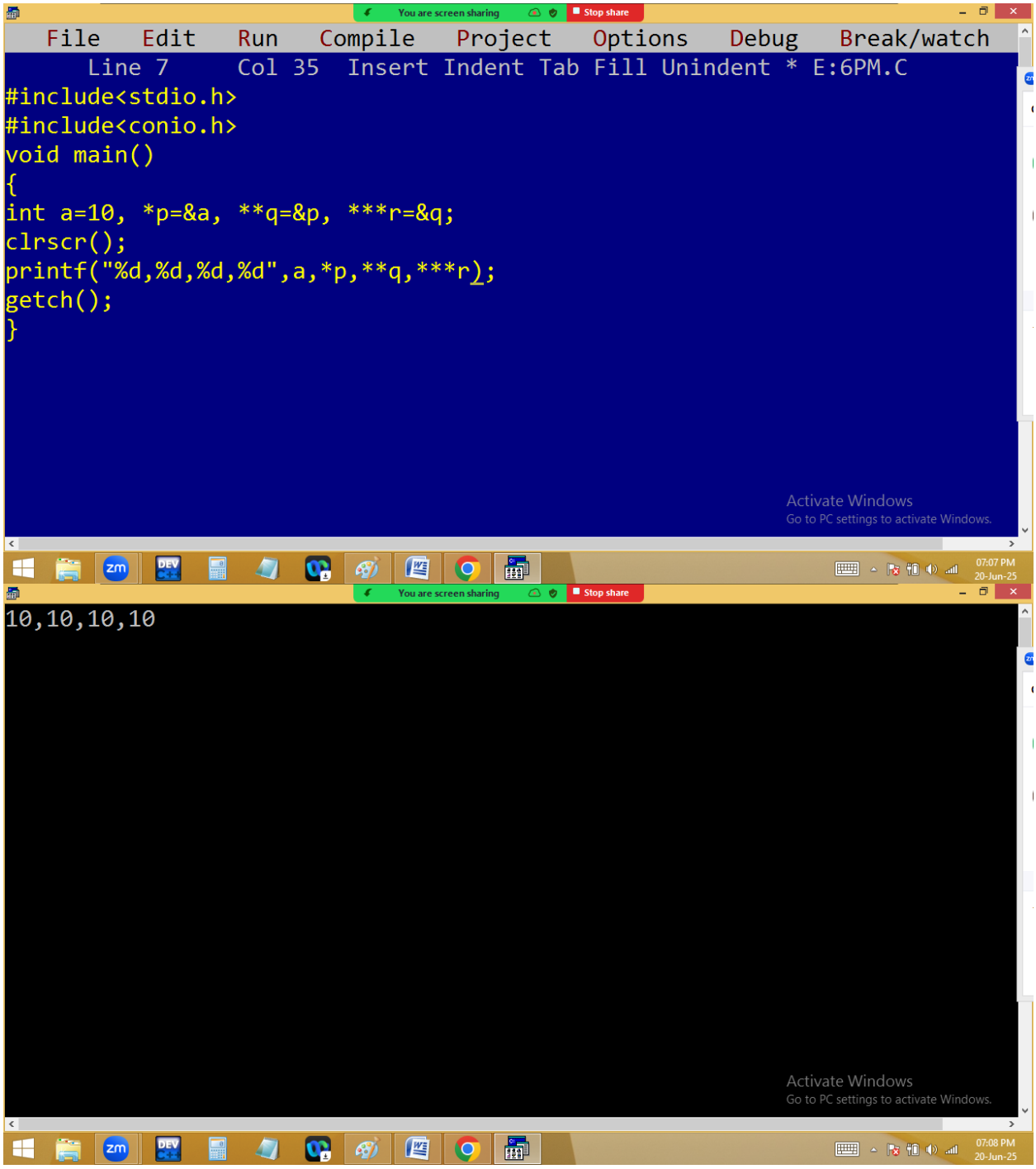
Pointer compatibility: Pointer can store only the same type of variable address. When we are storing different type of variable address, it gives garbage or runtime error.

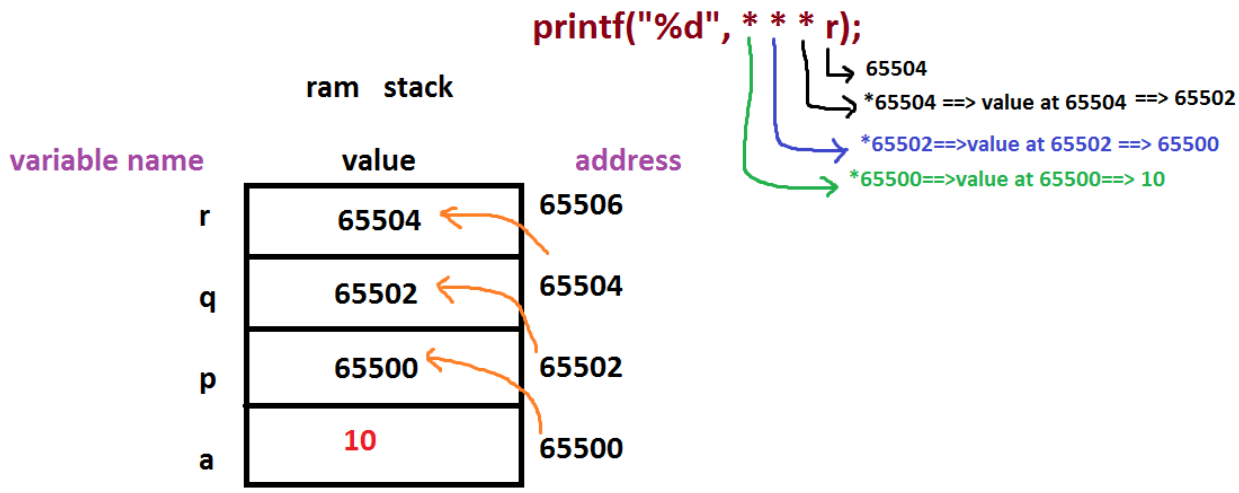

You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 1 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
float *p=&a;
clrscr();
printf("a=%d",*p);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:55 PM
20-Jun-25
You are screen sharing
Stop share
a=0
Activate Windows
Go to PC settings to activate Windows.
06:55 PM
20-Jun-25


You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 13 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
float a=10.5;
int *p=&a;
clrscr();
printf("a=%f",*p);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:56 PM
20-Jun-25
You are screen sharing
Stop share
printf : floating point formats not linked
Abnormal program termination
Activate Windows
Go to PC settings to activate Windows.
06:56 PM
20-Jun-25

Double pointer or pointer to pointer:

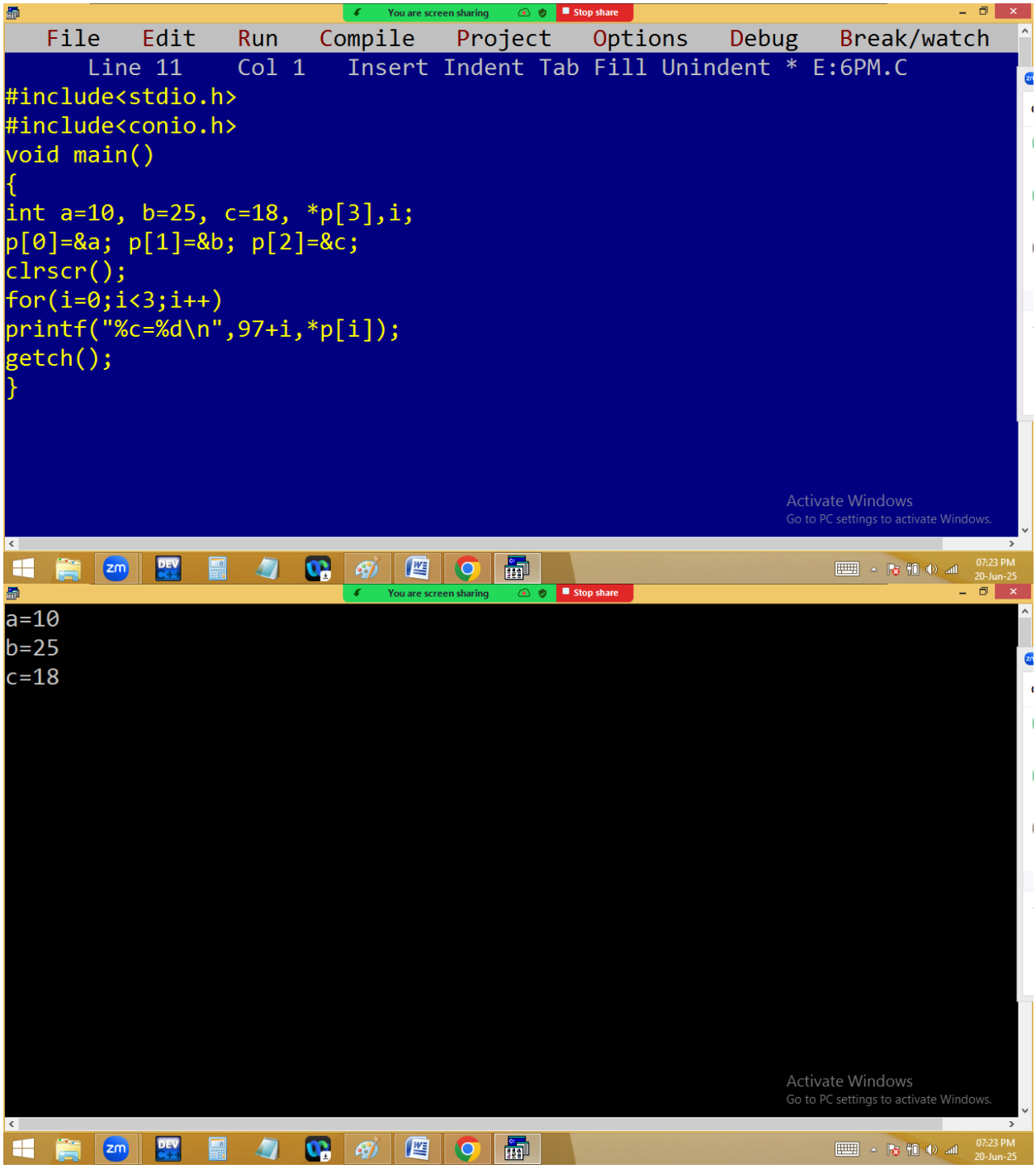
The pointer which stores the address of another pointer is called double pointer. They are used to handle dynamic multi-dimensional arrays.


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Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 35 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10, *p=&a, **q=&p, ***r=&q;
clrscr();
printf("%d,%d,%d,%d",a,*p,**q,***r);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:07 PM
20-Jun-25
You are screen sharing
Stop share
10,10,10,10
Activate Windows
Go to PC settings to activate Windows.
07:08 PM
20-Jun-25



Array of pointer: Like normal variables we can declare pointer using array. Due to this one pointer can several variable addresses. It reduce no of pointer declarations and complexity.

They are used in dynamic memory allocation to handle multi dimensional arrays.


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Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 1 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10, b=25, c=18, *p[3],i;
p[0]=&a; p[1]=&b; p[2]=&c;
clrscr();
for(i=0;i<3;i++)
printf("%c=%d\n",97+i,*p[i]);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:23 PM
20-Jun-25
You are screen sharing
Stop share
a=10
b=25
c=18
Activate Windows
Go to PC settings to activate Windows.
07:23 PM
20-Jun-25